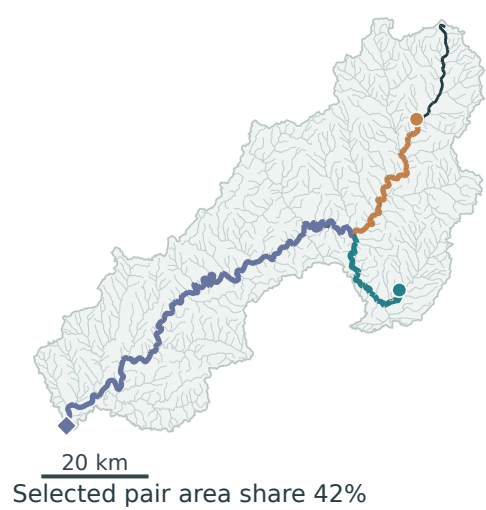


Real river geometry: where does matched storage act?

--- Translation control — Early branch — Late branch — Shared trunk

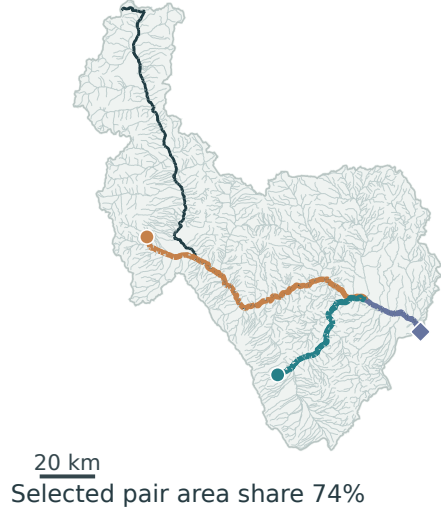
a Elongated / tributary-rich
03030852 · n=102



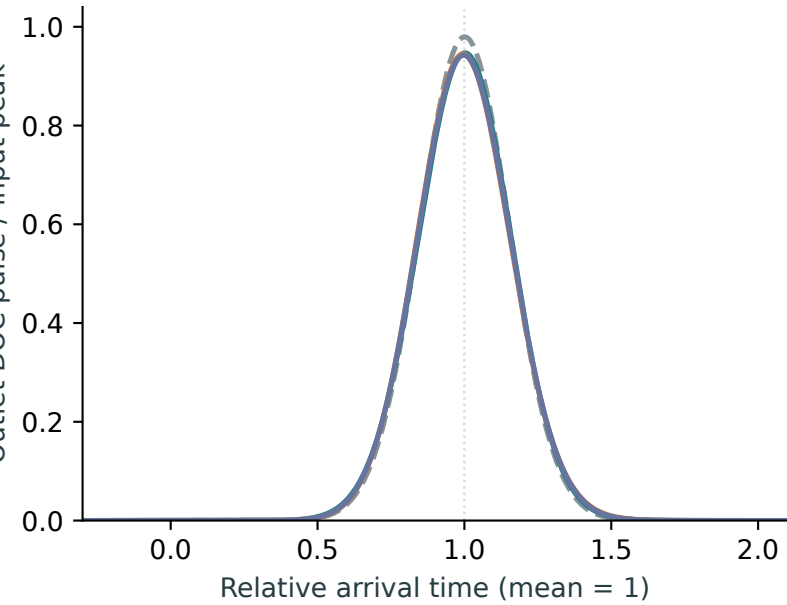
b Mainstem dominated / sparse
06893990 · n=28



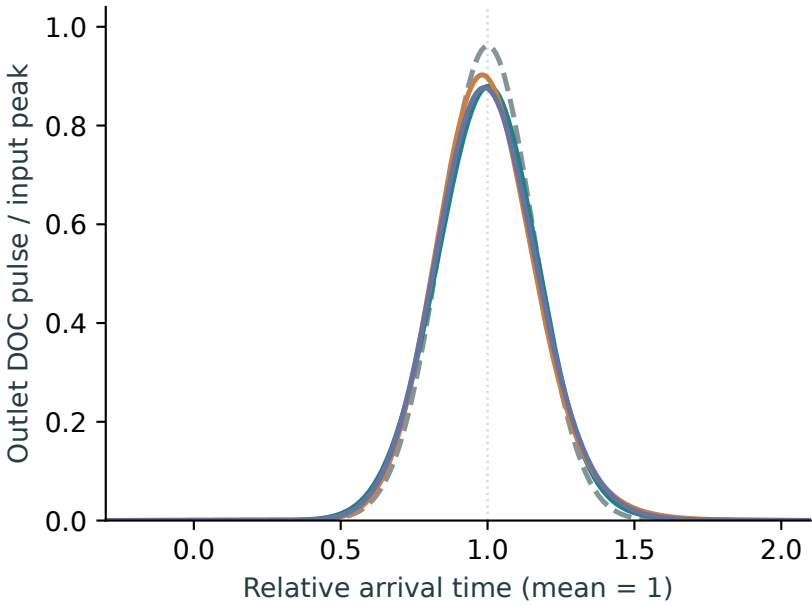
c Broad / tributary-rich
07099200 · n=165



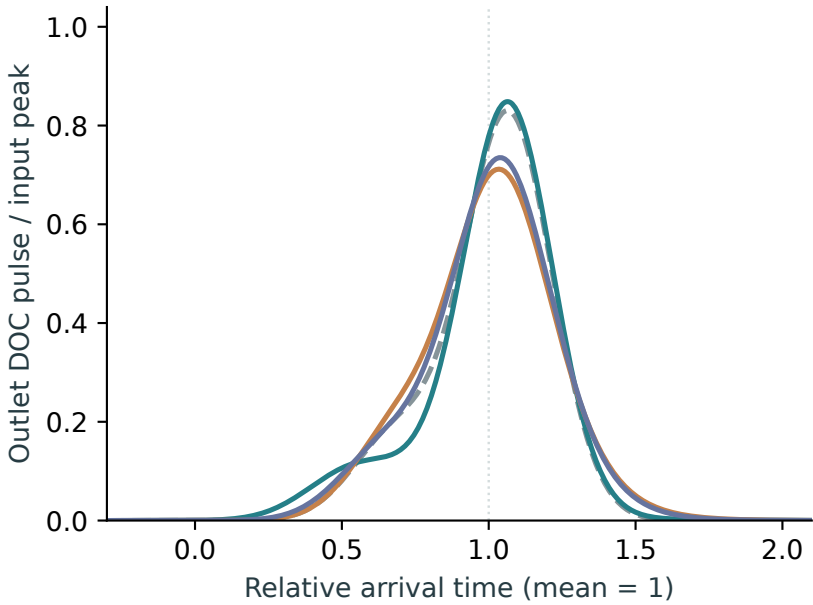
d Equal added variance, different placement



e Equal added variance, different placement



f Equal added variance, different placement



Actual NHDPlus flowlines; maps use independent physical scales. Identical forcing at two real representative branch locations. Controlled storage locations, not relocated mapped waterbodies. Fixed path means, unit gain and added variance; relative time is not measured days.